THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Two angles of a linear pair differ by 40°. The smaller angle is: 1

(a) 70° **(b)** 80° **(c)** 60° **(d)** 90°

2. For every whole number n, $n + (n+1) + (n+2)$ is always a multiple of: 1

(a) 2 **(b)** 6 **(c)** 3 **(d)** 4

3. Sides 13 cm, 6 cm and 8 cm are given. Which statement is true? 1

(a) only a right triangle is possible **(b)** only an equilateral triangle is possible

(c) no triangle can be formed **(d)** a triangle can be formed

4. $\frac{5}{7}$ of $2\frac{4}{5}$ is: 1

(a) $3\frac{18}{35}$ **(b)** 2 **(c)** 4 **(d)** 1

5. If 1 lakh = 1,00,000, how many lakhs are there in 3,40,00,000? 1

(a) 350 **(b)** 340 **(c)** 341 **(d)** 339

6. A cyclist covers 28 km in the first hour and 3 km less every following hour for 4 hours in all. The total distance (km) is: 1

(a) 99 **(b)** 94 **(c)** 93 **(d)** 95

7. What must be added to 0.875 to get 2? 1

(a) 1.025 **(b)** 1.225 **(c)** 1.125 **(d)** 2.125

8. If $x = 3$ and $y = 6$, the value of $5x - 4y + xy$ is: 1

(a) 8 **(b)** 11 **(c)** 10 **(d)** 9

9. One angle of a linear pair is 3 times the other. The larger angle is: 1

(a) 145° **(b)** 135° **(c)** 155° **(d)** 125°

10. If $1 + 2 + 3 + \cdots + n = 406$, then n is: 1

(a) 29 **(b)** 28 **(c)** 30 **(d)** 27

11. In an isosceles triangle with perimeter 50 cm the equal sides are 18 cm each. The base (cm) is: 1

(a) 15 **(b)** 13 **(c)** 14 **(d)** 16

12. If $x \times \frac{2}{5} = \frac{4}{15}$, then $x \div \frac{2}{3}$ is: 1

(a) $\frac{1}{2}$ **(b)** $1\frac{1}{2}$ **(c)** 1 **(d)** 2

13. The number of digits in the product 99,999 × 9,999 is: 1

(a) 11 **(b)** 10 **(c)** 8 **(d)** 9

14. The value of $(100 - (4 \times (12 - 7) + 10)) \div (2 \times 5)$ is: 1

(a) 5 **(b)** 70 **(c)** 7 **(d)** 9

15. Which fraction is equal to 0.375? 1

(a) $\frac{3}{7}$ **(b)** $\frac{3}{5}$ **(c)** $\frac{3}{8}$ **(d)** $\frac{3}{80}$

16. A rule is: output $= 3\times$ input -7. The output for the input 12 is: 1

(a) 28 **(b)** 30 **(c)** 31 **(d)** 29

17. An angle is equal to its supplement. The angle is: 1

(a) 180° **(b)** 45° **(c)** 90° **(d)** 60°

18. The number of reverse-and-add steps for 79 to become a palindrome is: 1

(a) 7 **(b)** 5 **(c)** 8 **(d)** 6

19. The two acute angles of a right triangle are in the ratio $2 : 3$. The smaller acute angle is: 1

(a) 26° **(b)** 46° **(c)** 36° **(d)** 56°

20. $\frac{2}{3} \div (\frac{4}{5} \div \frac{2}{5})$ equals: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{6}$ **(c)** $2\frac{1}{12}$ **(d)** $\frac{2}{3}$

21. How many lakhs are there in 35 crore? 1

(a) 3510 **(b)** 3500 **(c)** 4500 **(d)** 3490

22. The value of $96 \div (4 \times (7 - 3)) + 5 \times (11 - 8)$ is: 1

(a) 23 **(b)** 212 **(c)** 21 **(d)** 1862

23. Which of these is the greatest? 1

(a) $\frac{3}{4}$ **(b)** 0.85 **(c)** 0.8 **(d)** $\frac{7}{8}$

24. If $x = 5$, the value of $3x^2 + 4$ is: 1

(a) 81 **(b)** 80 **(c)** 79 **(d)** 78

25. Two supplementary angles are in the ratio $4 : 5$. The larger angle is: 1

(a) 110° **(b)** 100° **(c)** 120° **(d)** 90°

26. Which expression is a multiple of 11 for every digit pair (a, b)? 1

(a) $10a + b$ **(b)** $(10a + b) + (10b + a)$ **(c)** $a - b$ **(d)** $(10a + b) - (10b + a)$

27. The number of isosceles triangles with whole-number sides and perimeter 15 cm (equilateral included) is: 1

(a) 6 **(b)** 4 **(c)** 5 **(d)** 3

28. $\frac{4}{9}$ of a number is 20. Three times the number is: 1

(a) $67\frac{1}{2}$ **(b)** $135\frac{1}{2}$ **(c)** 270 **(d)** 135

29. The number of seconds in 100 days is: 1

(a) 86,36,400 **(b)** 86,40,000 **(c)** 86,41,000 **(d)** 8,64,000

30. The value of $48 \div (2 \times (5 + 1)) \times (14 - 6)$ is: 1

(a) 128 **(b)** 32 **(c)** 1688 **(d)** 34

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31. $0.5 \times 0.5 \times 0.8 =$ 1

(a) 2 **(b)** 0.2 **(c)** 20 **(d)** 0.02

32. If $a = 5$ and $b = 2$, the value of $2a^2 - 9b$ is: 1

(a) 31 **(b)** 33 **(c)** 34 **(d)** 32

33. The supplement of the complement of 50° is: 1

(a) 140° **(b)** 160° **(c)** 130° **(d)** 150°

34. The smallest 5-digit number divisible by 9 is: 1

(a) 10009 **(b)** 10007 **(c)** 10008 **(d)** 10017

35. The number of triangles with whole-number sides (in cm) and perimeter 14 cm is: 1

(a) 3 **(b)** 6 **(c)** 5 **(d)** 4

36. A pipe fills $\frac{5}{6}$ of a tank in 20 minutes. The time to fill the whole tank (minutes) is: 1

(a) $24\frac{1}{2}$ **(b)** 12 **(c)** 48 **(d)** 24

37. The number of zeros in 4,500 × 2,000 is: 1

(a) 7 **(b)** 8 **(c)** 5 **(d)** 6

38. The value of 'twice the sum of 15 and 4, decreased by 11' is: 1

(a) 38 **(b)** 27 **(c)** 16 **(d)** 49

39. How many times does 0.04 go into 1.2? 1

(a) 29 **(b)** 31 **(c)** 30 **(d)** 32

40. Simplify $6(x + 1) - 2(x - 4)$: 1

(a) $4x + 2$ **(b)** $8x + 14$ **(c)** $4x + 14$ **(d)** $4x - 2$

41. The reflex angle that goes with an angle of 100° (so that together they make a complete turn) is: 1

(a) 260° **(b)** 270° **(c)** 250° **(d)** 280°

42. The largest 4-digit number with all digits different that is divisible by 10 is: 1

(a) 9870 **(b)** 9871 **(c)** 9872 **(d)** 9869

43. The exterior angle at the base of an isosceles triangle is 140°. The apex angle is: 1

(a) 110° **(b)** 120° **(c)** 100° **(d)** 90°

44. $\frac{3}{8}$ of a wall is already painted. Then $\frac{3}{5}$ of the remaining part is painted. The fraction of the whole wall still unpainted is: 1

(a) $\frac{1}{8}$ **(b)** $\frac{9}{20}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{2}$

45. 1 crore = 1,00,00,000; the number of crores in 80,00,000 is: 1

(a) 0.08 **(b)** 80 **(c)** 8 **(d)** 0.8

46. Compare 17 × 11 and 16 × 12. The first minus the second is: 1

(a) -5 **(b)** -4 **(c)** -6 **(d)** -3

47. 0.25 of a number is 3.5. The number is: 1

(a) 13 **(b)** 14 **(c)** 16 **(d)** 15

48. The perimeter of a rectangle with sides $4x + 9$ and $6x - 5$ is: 1

(a) $-4x + 28$ **(b)** $48x$ **(c)** $20x + 8$ **(d)** $10x + 4$

49. Two complementary angles differ by 10°. The larger angle is: 1

(a) 60° **(b)** 50° **(c)** 70° **(d)** 40°

50. Kaprekar's routine on 1459 reaches 6174 after how many steps? 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 5

END OF QUESTION PAPER **Total: 50 Marks**

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